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A new displacement demodulation algorithm for the phase-modulated self-mixing interferometer based on FPGA

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Abstract

This paper introduces a design of the phase demodulation system based on FPGA for the phase-modulated self-mixing interferometer. By the processing that the modulation SIN signal and the doubling frequency modulation COS signal are respectively multiplied by the phase-modulated self-mixing signal(PSMI) and then low-pass filtered and shaped in the electrical domain , two orthogonal square wave signals sensitive to the object's movement direction are obtained for displacement reconstruction. The FPGA signal processing system controls the generation of a highly stable and accurate sinusoidal signal source for the modulation of the electro-optic crystal(EOM). At the same time, the two orthogonal square wave signals obtained after the demodulation processing are subdivided for the high-precision measurement of a uniformly moving object's displacement. The high-frequency amplifier circuit amplifying the self-mixing interference signal, multiplier & frequency doubler circuits based on MLT04 and low-pass filter circuits are designed. FPGA-based counting programs , DAC circuits and programs used for generating sine waves are also designed. In the experiment, comparison tests for displacement measurement are performed using the PI company's two-dimensional precision electronically-controlled translation stage, and high-precision displacement measurement can be achieved. The demodulation system does not rely on a PC and a high-precision acquisition card, and can realize real-time measurement of displacement, greatly improving the practicability of the system.

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1. Preface

Since the laser's self-mixing effect was discovered many years ago, it is widely applied to engineering fields, such as displacement measurement, vibration measurement ^[1-3], and absolute distance measurement, due to its advantages of simpler optical system construction than traditional optical interference. At the same time, researchers have found self-mixing interference effects in various lasers, such as DFB lasers ^[4], He-Ne lasers, and fiber lasers. Many scholars have also conducted many in-depth and systematic researches on phase-modulated laser self-mixing interference ^[5-8]. For the demodulation methods, there are FFT frequency-domain transforming demodulation method and time-domain orthogonal demodulation method. For the perspective of improving the engineering practicality of phase-modulated laser self-mixing interference, this paper proposes a demodulation algorithm using FPGA signal processing system to reconstruct the displacement when objects in the external cavity moves at a uniform speed. The demodulation hardware system does not depend on the PC and the expensive acquisition card. In the 100 mm displacement measurement experiment result, it has achieved an accuracy of 0.7μm, which has strong practicability in many fields.

2. Principle of the electrical domain demodulation algorithm to extract the amplitude of the first and second harmonics

It is assumed that the laser is modulated sinusoidally by an electro-optic crystal placed in the outer laser cavity ^[9], where the modulation function is $\zeta(t) = b \sin(\omega_p t + \gamma)$, b is the modulation depth, ω_p is the modulation angular frequency $\omega_p = 2\pi f_p$, and γ is the initial phase of the modulation. Because the light passes through the electro-optic crystal twice in the external cavity, the phase change caused by the electro-optic crystal modulation is $2\zeta(t)$. The self-mixing interference system is adjusted to work under weak feedback conditions, and the multiple feedback effects of light in the external cavity are ignored. The self-mixing interference light intensity signal can be expressed as:

$$Q(t) = Q_0 \{1 + p \cos[\sigma(t) + 2b \sin(\omega_p t + \gamma)]\} \quad (2-1)$$

Expanding (2-1) by Bessel functions, we get $Q(t)$ as follows:

$$\begin{aligned} Q(t) = & Q_0 [1 + p \cos \sigma(t) J_0(2b)] \\ & + 2pQ_0 \cos \sigma(t) \sum_{n=1}^{\infty} J_{2n}(2b) \cos[(2n)(\omega_p t + \gamma)] \\ & - 2pQ_0 \sin \sigma(t) \sum_{n=1}^{\infty} J_{2n-1}(2b) \sin[(2n-1)(\omega_p t + \gamma)] \end{aligned} \quad (2-2)$$

$J_{2n}(2b)$ and $J_{2n-1}(2b)$ are the even and odd order Bessel functions respectively, and the first and second harmonic components can be found by the light intensity signal. The expression can be described as (2-3) and (2-4) :

$$\begin{aligned} Q(\omega_p, t) = & -2pQ_0 \sin \sigma(t) J_1(2b) \sin(\omega_p t + \gamma) \\ = & B_1(t) \sin(\omega_p t + \gamma) \end{aligned} \quad (2-3)$$

$$\begin{aligned} Q(2\omega_p, t) = & 2pQ_0 \cos \sigma(t) J_2(2b) \cos(2\omega_p t + 2\gamma) \\ = & B_2(t) \cos(2\omega_p t + 2\gamma) \end{aligned} \quad (2-4)$$

$B_1(t)$ and $B_2(t)$, respectively, are amplitudes of the first and second harmonic component, $B_1(t)$ which are subject to the sine modulation of the phase $\sigma(t)$ and $B_2(t)$ the cosine modulation of the phase $\sigma(t)$. The detailed expressions of $B_2(t)$, $B_1(t)$ are as follows:

$$B_1(t) = -2pQ_0J_1(2b)\sin\sigma(t) \quad (2-5)$$

$$B_2(t) = 2pQ_0J_2(2b)\cos\sigma(t) \quad (2-6)$$

The phase information $\sigma(t)$ can be demodulated from $B_1(t)$, $B_2(t)$ by arctangent operation. The specific method is as follows:

$$\tan\sigma(t) = -\frac{J_2(2b)}{J_1(2b)} \cdot \frac{B_1(t)}{B_2(t)} \quad (2-7)$$

$$\sigma(t) = \arctan\left[-\frac{J_2(2b)}{J_1(2b)} \cdot \frac{B_1(t)}{B_2(t)}\right] \quad (2-8)$$

The phase demodulated by the above algorithm is wrapped in the range of $-\pi$ and π . For obtaining the true phase value $\sigma(t)$, the unwrapping process will be performed. On the basis of the mathematical relationship between the phase and the length of the external cavity of the self-mixing interferometer, the external reflection object's displacement can be recovered as shown in the following formula (2-9):

$$l(t) = \frac{\sigma(t)}{4\pi\eta} \lambda_0 \quad (2-9)$$

λ_0 is the laser's wavelength, while η is the air's refractive index and $l(t)$ is the displacement of the externally reflecting object.

Therefore, as long as the amplitude of the first harmonic component $B_1(t)$ and the second harmonic component $B_2(t)$ of the laser self-mixing interference light intensity signal is obtained, the object's displacement can be obtained. The principle of the displacement demodulation algorithm proposed in this article is shown in Figure 1 below:

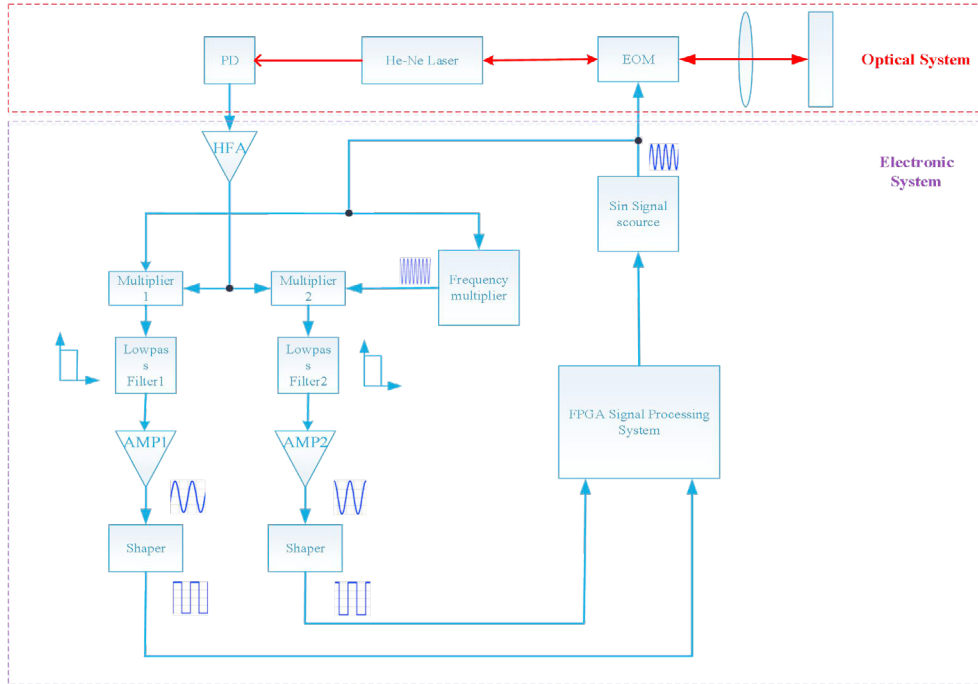


Fig. 1. Principle of electric domain demodulation algorithm for extracting the amplitude of the first and second harmonics

(1) A high-quality sine wave can be generated by using the FPGA to control the DAC and then being filtered. The sine wave is used to modulate the electro-optic crystal (EOM);

(2) By multiplying the sine wave signal and the phase-modulated self-mixing interference signal (PSMI), the first order harmonic component is converted to the zero frequency, and then we can obtain the first order harmonic component's amplitude $B_1(t)$ by a low pass filter. Specifically:

$$\begin{aligned}
 Z(t) &= Q(t) \cdot \zeta(t) \\
 &= \{Q_0[1 + p \cos \sigma(t) J_0(2b)] + 2pQ_0 \cos \sigma(t) \sum_{n=1}^{\infty} J_{2n}(2b) \cos[(2n)(\omega_p t + \gamma)] \\
 &\quad - 2pQ_0 \sin \sigma(t) \sum_{n=1}^{\infty} J_{2n-1}(2b) \sin[(2n-1)(\omega_p t + \gamma)]\} \cdot b \sin(\omega_m t + \gamma)
 \end{aligned} \tag{2-10}$$

From the formula(2-10), some conclusions can be obtained. When the phase-modulated self-mixing interference signal (PSMI) is multiplied with a sinusoidal carrier, from the frequency domain perspective, it can be found that the original DC term and the zeroth harmonic component are transformed to the frequency ω_p . And the first order harmonic component is converted to the zero frequency after being multiplied by the sine modulating wave. The result is as follows:

$$\begin{aligned}
 Q_1(\omega_p, t) &= Q(\omega_p, t) \cdot \zeta(t) = -2pQ_0 \sin \sigma(t) J_1(2b) \sin(\omega_p t + \gamma) \cdot b \sin(\omega_p t + \gamma) \\
 &= \frac{1}{2} b B_1(t) - \frac{1}{2} b B_1(t) \cos(2\omega_p t + 2\gamma)
 \end{aligned} \tag{2-11}$$

The second order and higher order harmonic components being multiplied with the sin modulation carrier wave is transformed $n\omega_p$ everywhere ($n = 1, 2, 3 \dots$). So a low pass filter can be used to obtain the first order harmonic component's amplitude $\frac{1}{2}bB_1(t)$, filtering out other frequency components;

(3) By multiplying the phase-modulated self-mixing interference signal (PSMI) with the frequency-doubled sine modulation signal, the second-order harmonic component's amplitude $B_2(t)$ will be obtained by a low-pass filter.

Specifically, the modulation signal $\zeta(t) = b \sin(\omega_p t + \gamma)$ is subjected to frequency doubling processing:

$$\zeta^2(t) = b^2 \sin^2(\omega_p t + \gamma) = -\frac{1}{2}b^2 \cos[2(\omega_p t + \gamma)] + \frac{1}{2}b^2 \quad (2-12)$$

After the DC blocking processing, an AC component or a frequency-doubled signal of the modulation signal is obtained.

$$\zeta_{ac}^2(t) = -\frac{1}{2}b^2 \cos[2(\omega_p t + \gamma)] \quad (2-13)$$

By multiplying $\zeta_{ac}^2(t)$ with the laser self-mixing interference signal (PSMI), we get

$$\begin{aligned} R(t) &= Q(t) \bullet \zeta_{ac}^2(t) \\ &= \{Q_0[1 + p \cos \sigma(t) J_0(2b)] + 2pQ_0 \cos \sigma(t) \sum_{n=1}^{\infty} J_{2n}(2b) \cos[(2n)(\omega_p t + \gamma)] \\ &\quad - 2pQ_0 \sin \sigma(t) \sum_{n=1}^{\infty} J_{2n-1}(2b) \sin[(2n-1)(\omega_p t + \gamma)]\} * \{-\frac{1}{2}b^2 \cos[2(\omega_p t + \gamma)]\} \end{aligned} \quad (2-14)$$

From the frequency domain perspective, it can be found that the original DC component and zero order harmonic component are converted to the frequency $2\omega_p$, and the second order harmonic component being multiplied by the doubled modulating carrier wave is converted to the zero frequency.

$$Q_2(2\omega_p, t) = Q(2\omega_p, t) * \zeta_{ac}^2(t) = \frac{1}{4}b^2 B_2(t) + \frac{1}{2}b^2 pQ_0 J_2(2b) \cos \sigma(t) \cos[4(\omega_p t + \gamma)] \quad (2-15)$$

And the 1st and 3rd or higher harmonic components which is multiplied with $\zeta_{ac}^2(t)$ is transformed to $n\omega_p$ ($n = 1, 2, 3 \dots$). So a low pass filter can be used to obtain the amplitude of the second-order harmonics $\frac{1}{4}b^2 B_2(t)$, filtering out other frequency components;

(4) When the object in the outer cavity moves in a straight line at a constant speed, under weak feedback conditions, the signal out from the laser self-mixing interferometer is a criterion sine wave, and its frequency f_d is proportional to the object's movement speed v . It is also called the Doppler frequency [10-11]:

$$f_d = 2 \frac{v}{\lambda_0} \quad (2-16)$$

Therefore, the external cavity's phase change caused by uniform motion, the demodulated amplitude of the first and second harmonics will be standard sine and cosine signals, and the phase difference is sensitive to the object's

movement direction. The sine and cosine signal is then shaped into a square wave, obtained and divided by an FPGA signal processing system to reconstructing the object's displacement.

It is worth noting that the first-order harmonic component's amplitude and the second-order harmonic component's amplitude are modulated by the first order and the second order Bessel function, independently, as shown in Figure 2 below. And the modulation depth b needs to avoid the zero positions of the first-order and second-order Bessel functions, otherwise the modulation is meaningless.

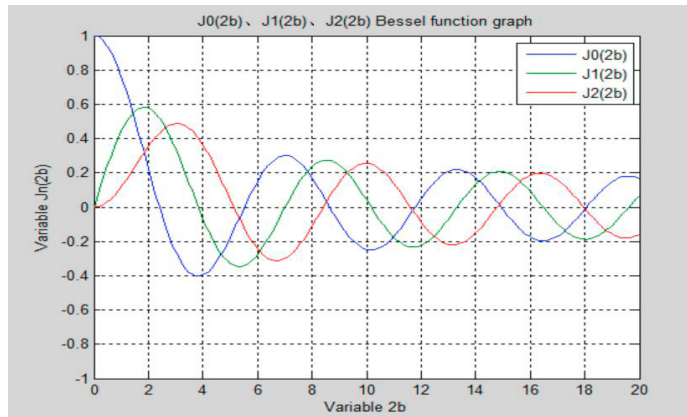


Fig. 2. Variations of zero-order, first-order, and second-order Bessel functions with modulation depth $2b$

3. Design of Displacement Demodulation Circuit System

It can be known from the above demodulation algorithm that a hardware circuit system needs to be designed in the electronic domain to perform the displacement reconstruction, which mainly includes the design of the following circuit sub-modules.

3.1. High frequency amplifier circuit

The high frequency amplifier circuit uses the TI company's OPA2614 operational amplifier, which has a high gain bandwidth of 290MHz, meeting the amplification need of the phase-modulated self-mixing interference signal.

3.2. Multiplier and Frequency Multiplier circuits based on MLT04

MLT04 is an analog multiplier. It is a voltage input and output type. It has four independent channels. It is a four-quadrant multiplier with a bandwidth of 8MHz. The voltage output is fixed. The typical linear error of the X or Y input is 0.2%. The super temperature stability is 0.005%. MLT04 is used to realize the multiplication operation of phase-modulated self-mixing interference signal (PSMI) and sinusoidal modulation carrier SINW, the frequency multiplication of the sinusoidal modulation carrier, and the multiplication operation of carrier signal COS2W and PSMI signal. The circuit schematic of the MLT04 is shown in Figure 3 below.

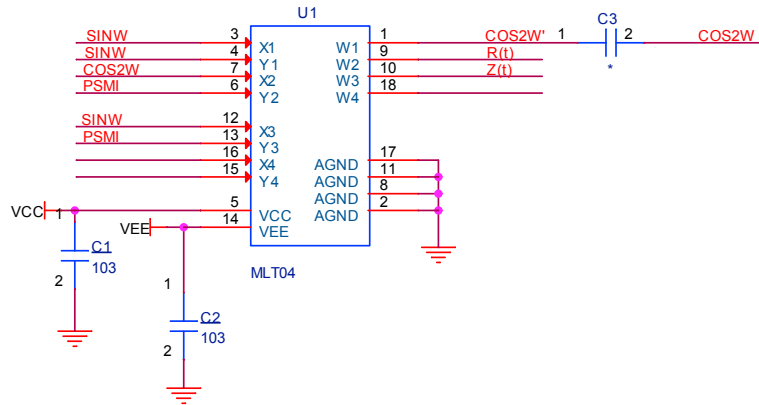


Fig. 3. Circuit diagram of frequency multiplier and multiplier using MLT04

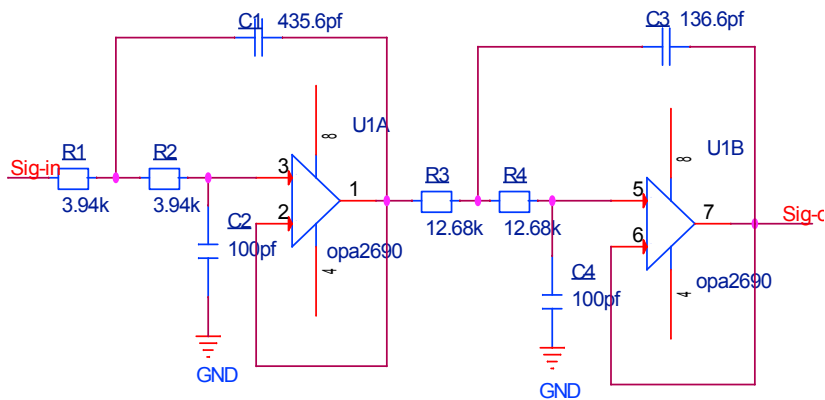


Fig. 4. Schematic of the low-pass filter circuit

3.3. Low-pass module

According to the specific parameters of the design, the TI's OPA2690 was used to design the low-pass filter as Figure 4 shows. The fourth-order Butterworth filter was designed using the FilterPro filter design software. The corresponding 3DB bandwidth frequency of the low-pass filter is 100KHZ. Candence software is used to design the actual circuit diagram and PCB, and the actual circuits is made. The test meets the expected function. After being connected to the hardware demodulation system, it can be found that two orthogonal signals sensitive to the object's motion direction can be obtained. When the object moves forward, the demodulated signal is shown in Figure 5 below. when the object moves in the reverse direction the demodulated signal is shown in Figure 6 below. It is consistent with the theoretical analysis.

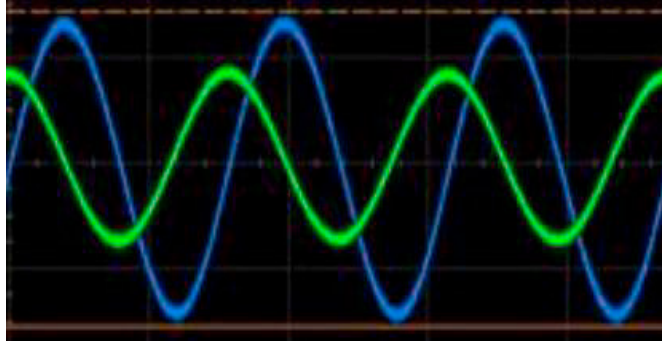


Fig. 5. Two signals obtained by demodulation when the object is moving forward

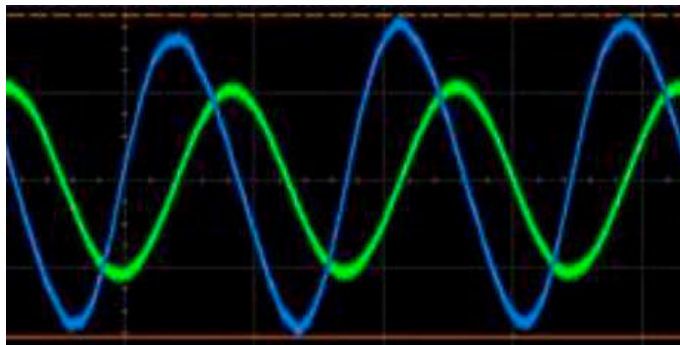


Fig. 6. Two signals obtained by demodulation when the object is moving backward

3.4. Signal shaping circuit

The signal shaping uses the zero crossing comparison circuit designed by the OPA series of operational amplifier. Two square wave signals are obtained by the shaping circuit. After giving an offset voltage, the signals are sent to the FPGA signal processing system to count and recover the object's displacement.

3.5. Sine wave signal generation and displacement reconstruction system based on FPGA

The FPGA signal processing system mainly is used to realize two functions. One is to generate a sine wave with adjustable, precise, stable frequency through a look-up table method. It is used for sinusoidal modulation of electro-optic crystal and multiplication with phase-modulated self-mixing interference signal. The second is used to count and divide the two demodulated orthogonal signals, reconstruct the displacement and display the real-time displacement value on the LCD liquid crystal as shown in Figure 7 below.

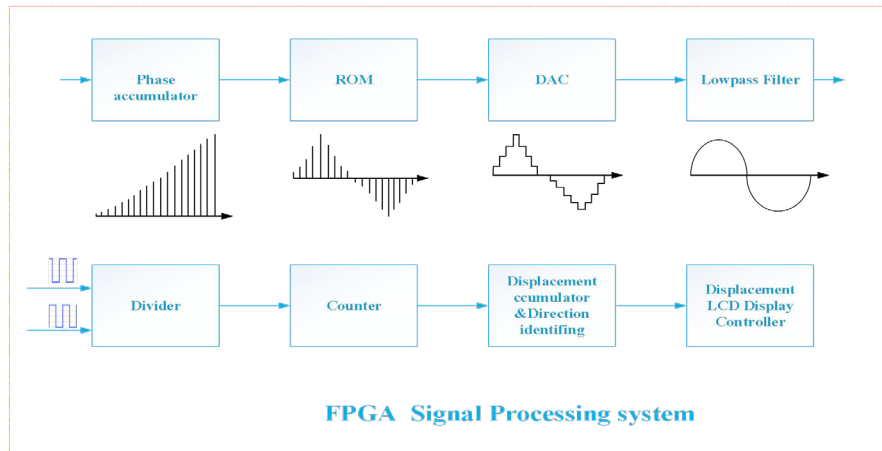


Fig. 7. The sinusoidal signal source generation and displacement reconstruction & display based on FPGA schematic diagram

4. Displacement measurement verification test

Experimental method: We use the PI company's two-dimensional precision electronically-controlled translation stage M-521.DD (DC motor driven, unidirectional repetitive positioning resolution: $0.1 \mu\text{m}$, maximum running speed : 50 mm/s, 200 mm range) to drive the movement of the target cat-eye lens^[12]. The electronic translation stage's motion range is set to 100mm, and the motion speed is set to 15mm/s. Every step is 10mm at a constant speed and the step is 10 times. The liquid crystal display screen reads the phase-modulated self-mixing interferometer's displacement measurement and real-time error for each step. Do lots of groups and save the experimental data. Four groups of the experimental data are randomly selected for analysis. Mea1, Mea2 and Mea3 were performed under the same measurement environmental conditions (constant temperature ($20 \text{ }^{\circ}\text{C} \pm 1 \text{ }^{\circ}\text{C}$), constant humidity ($50\% \pm 3\%$), laser preheating for 4 hours, laser wavelength 632.8334nm), and Mea4 was performed from another test site.

Data processing: four groups' data (Mea1, Mea2, Mea3, Mea4) obtained in the experiments are as Table 4.1 shows:

Table 4.1. Experimental data PI rail displacement: PI Dis; Measured displacement: Mea; Error: Err; Unit: μm

PI Dis	0	10000	20000	30000	40000	50000	60000	70000	80000	90000	100000
Mea1	0	9999.7	19999.5	29999.4	39999.6	49999.3	60000.3	69999.4	79999.6	90000.6	99999.3
Err 1	0	-0.3	-0.5	-0.6	-0.4	-0.7	0.3	-0.6	-0.4	0.6	-0.7
Mea2	0	9999.7	19999.4	29999.6	39999.6	49999.5	59999.8	69999.3	80000.4	90000.3	100000.5
Err 2	0	-0.3	-0.6	-0.4	-0.4	-0.5	-0.2	-0.7	0.4	0.3	0.5
Mea3	0	9999.5	20000.2	29999.6	39999.7	49999.4	59999.7	69999.8	79999.6	90000.3	99999.5
Err 3	0	-0.5	0.2	-0.4	-0.3	-0.6	-0.3	-0.2	-0.4	0.3	-0.5
Mea4	0	10000	19999.5	30000.7	40000.6	49999.6	60000.5	70000.3	79999.7	89999.8	99999.6
Err 4	0	0	-0.5	0.7	0.6	-0.4	0.5	0.3	-0.3	-0.2	-0.4

It can be known from Table 3.1 that the maximum error of the interferometer relative to the PI's electronic translation stage measurement is $0.7 \mu\text{m}$, which indicates that the interferometer has the function of real-time measurement in the large-scale displacement of a hundred millimeters, and has obtained a less than $0.7 \mu\text{m}$ measurement error. The measurement repeatability and reproducibility is better.

5. Conclusion

This paper proposes a method of displacement demodulation and real-time reconstruction using the method of extracting the first-harmonic's and second-harmonic's amplitude of the phase-modulated self-mixing interference signal (PSMI) in the electrical domain when the object in the outer self-mixing laser cavity moves at a uniform speed. The core of signal processing is implemented by the FPGA system, and an electric-domain phase-modulated self-mixing demodulation circuit system is designed, including a high-frequency amplifier circuit, a frequency multiplication circuit and two multiplier circuits based on MLT04, low-pass filter circuits, etc. In the experiment, a comparative experiment of displacement measurement was performed using the PI company's two-dimensional precision electronically-controlled translation stage. The experimental data results show that in the 100 mm displacement range, a measurement accuracy of 0.7μm can be achieved. The displacement demodulation and reconstruction system does not rely on a PC and a high-precision acquisition card, and has certain practicability.

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